Reply to the report of Referee 1

Manuscript: *Chiral soliton lattice in inhomogeneous magnetic fields*
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We thank the referee for a careful reading of our manuscript, for the very positive assessment, and for the questions, which helped us improve the presentation and add physical content to the paper. Below we reply to all comments point by point. The referee's comments are reproduced in italics. Equation and figure numbers refer to the revised manuscript unless stated otherwise; note that one figure has been added (new Fig. 2), so that the subsequent figures are renumbered. All changes are highlighted in the accompanying marked-up version of the manuscript, with the exception of the notational change described under point (1), which affects too many places to be usefully highlighted.

Point (1): notation $\boldsymbol{H}$

Isn't the notation H unfortunate because it usually refers to the auxiliary magnetic field in the presence of a medium? Although there is a medium here I think H does not have this usual meaning, which may cause a slight, but unnecessary, confusion.

Reply: We agree. Our $\boldsymbol{H}$ was simply the external magnetic field rescaled by a constant, $\mu\boldsymbol{B}/(4\pi^2 f_\pi^2)$, and has nothing to do with the auxiliary field of magnetostatics. We have therefore renamed it to $\boldsymbol{b}$ throughout the paper, including all derived quantities ($\bar{\boldsymbol{b}}$, $\bar{b}_{\mathrm{CSL}}$, $\bar{b}_{\mathrm{cr}}$, b_0, b_{DW}, b_ξ, etc.).

Changes in the manuscript: The symbol $\boldsymbol{H} \to \boldsymbol{b}$ has been replaced everywhere in the text, equations and figure captions. Below Eq. (3), we have added an explicit remark that $\boldsymbol{b}$ is merely the rescaled external magnetic field and must not be confused with the auxiliary field $\boldsymbol{H}$. The axis label of Fig. 1 has been updated accordingly. While doing so, we have also noticed that the stream function of the Gaussian magnetic field in Eq. (32) was denoted by the same letter ψ as the scalar potential $\psi[\boldsymbol{b}]$ of the gradient–solenoid decomposition (18), although the two are unrelated. We have renamed the former to χ.

Point (2): finite-volume phase transition and Fig. 1

I am confused by the discussion at the end of Sec 2 and by Fig 1. Eqs (14) and (15) give two different classes of solutions with ϕ nonzero. Then there is the third possibility of $\phi = 0$. Why is the transition between the former two identified with the analog of the CSL transition? Does that mean in Fig 1 there is a $\phi \neq 0$ state everywhere, below and above the curve? I suspect I am misunderstanding the explanation, but since other readers might have the same problem, a clarification would be useful.

Reply: Indeed, in a finite volume, the ground state will always have nonzero ϕ, as required by the natural boundary condition (17) [formerly (16)]. Namely, $\phi = 0$ gives $\boldsymbol{n} \cdot \boldsymbol{\nabla}\phi = 0$, whereas Eq. (17) requires $\boldsymbol{n} \cdot \boldsymbol{\nabla}\phi = \boldsymbol{n} \cdot \boldsymbol{b} \neq 0$, so the trivial vacuum is not even a stationary point of the energy functional. As a consequence, the ground state will have negative total energy for any (nonzero) magnetic field and any choice of the system size. To answer the referee's question directly: yes, there is a $\phi \neq 0$ state everywhere in Fig. 1, both below and above the curve. We therefore need a different criterion to locate the phase transition than a simple comparison of the total energy to that of the trivial vacuum ($\phi = 0$) state.

The criterion we use is whether or not the ground state contains a soliton. The two classes of solutions (15) and (16) [formerly (14) and (15)] differ by the value of the pion field at the center of the domain:

- Below the curve in Fig. 1, the ground state is of the type (16), with $\phi_0(0) = 0$. The field stays close to the vacuum value in the bulk and only bends in a layer of thickness $\sim 1/m_\pi$ near the boundary in order to satisfy Eq. (17). The energy of this deformation is a surface effect that does not contribute to the energy density as $\bar{L} \to \infty$. This state is the finite-volume counterpart of the trivial vacuum.
- Above the curve, the ground state is of the type (15), with $\phi_0(0) = \pi$. The center of the domain now hosts a soliton, across which the pion field winds through the maximum of the potential. This is the finite-volume analog of the single domain wall that signals the onset of the CSL phase in infinite volume.

The curve in Fig. 1 thus marks the magnetic field at which it first becomes energetically favorable for a soliton to enter the system. In the limit $\bar{L} \to \infty$, the boundary layers become irrelevant, and the curve approaches the infinite-volume critical field $\bar{b}_{\rm CSL} = 4/\pi$. To make this explicit, we have added a new figure showing the ground-state profiles just below and just above the transition for $\bar{L} = 10$.

Changes in the manuscript: At the end of Sec. 2.2, we have added a paragraph explaining that $\phi = 0$ violates the natural boundary condition, so that $\phi_0 \neq 0$ on both sides of the curve in Fig. 1. The subsequent paragraph has been rewritten to explain the physical meaning of the two classes of solutions and of the curve in Fig. 1. We have added a new Fig. 2 showing the pion field and its gradient just below and just above the transition, and we have expanded the caption of Fig. 1. We have also corrected two minor typos in this subsection: $2L \to 2\bar{L}$ in Eq. (14), and $\partial_z \phi_0 = H \to \partial_{\bar{z}} \phi_0 = \bar{b}$ in the text below Eq. (17).

Point (3): physical lessons from Figs. 3 and 4

Are there any physical lessons to be learned from the obtained configurations, say in Figs 3 and 4? I understand that the focus of the paper is mostly on developing the mathematical tools, but the reader might wonder if the behavior of the baryon number is expected or surprising etc. Some physical explanation/intuition would then also be useful to predict the qualitative behavior of configurations with more complicated inhomogeneities in the magnetic field.

Reply: We thank the referee for this suggestion, which prompted us to analyze the obtained configurations further. Both figures can in fact be understood analytically, and they illustrate a few general mechanisms. We have added the following discussion to the manuscript.

(a) *Frustration.* The anomalous term favors alignment of $\boldsymbol{\nabla}\phi$ with $\boldsymbol{b}$, but a gradient has vanishing circulation. The pion field therefore cannot follow the magnetic field wherever the field lines circulate or reverse direction. The gradient–solenoid decomposition makes this precise: the ground state locks onto the gradient part $\boldsymbol{\nabla}\psi[\boldsymbol{b}]$, which is fixed by the flux of $\boldsymbol{b}$ through the boundary, whereas closed field loops inside the system are invisible to it. Since $\boldsymbol{\nabla} \times \boldsymbol{B}$ is proportional to the electric current density for static fields, fields in current-free regions are conservative and thus ideal for a CSL.

(b) *Baryon number.* In the chiral limit, the EoM and boundary condition imply $\int_V \boldsymbol{b} \cdot \boldsymbol{\nabla}\phi_0 = \int_V (\boldsymbol{\nabla}\phi_0)^2$. Hence the total baryon number of the ground state is

$$N_{\rm B} = \frac{f_\pi^2}{\mu} \int_V (\boldsymbol{\nabla}\phi_0)^2 = -\frac{2E[\phi_0]}{\mu} \geq 0$$

[new Eq. (22)]. Thus, any reduction of the condensation energy due to the inhomogeneity is accompanied by the same relative reduction of the baryon number. The local baryon number density, on the other hand, can be negative.

(c) *Fig. 3 (Gaussian field).* The field lines are circles centered at the origin. On a disk, the field would be tangential to the boundary everywhere, and the ground state would be exactly the trivial vacuum. On the square, the CSL is driven only by the boundary flux, which is suppressed by $e^{-L^2/(4R^2)} \approx 2 \times 10^{-3}$. Accordingly, the baryon number density is tiny: its maximum is of order 10^{-4} times $\max \boldsymbol{b}^2$, the density that a uniform field of the same strength would induce. The pattern can be derived analytically. The boundary data are invariant under 90° rotations and odd under reflections across the axes, which selects the leading harmonic $\phi_0 \propto xz(x^2 - z^2) \propto r^4 \sin 4\theta$. This gives $\boldsymbol{b} \cdot \boldsymbol{\nabla}\phi_0 \propto -r^4 e^{-r^2/R^2} \cos 4\theta$: negative on the axes and positive on the diagonals, with extrema at $r = \sqrt{2}R \approx 2.8$, exactly as in Fig. 3. The total baryon number is positive but quadratic in the small boundary flux. The lesson is that magnetic fields whose lines close inside the system are very inefficient at inducing a CSL.

(d) *Fig. 4 (domain wall).* The pion field would like to behave as $\pm b_0 z$ on the two sides of the wall. These profiles cannot be joined, and the odd solution vanishes on the wall, and so does the baryon density. The maximum principle applied to the harmonic function $\partial_z \phi_{\rm DW}$ shows that $0 \leq \boldsymbol{b} \cdot \boldsymbol{\nabla}\phi_{\rm DW} \leq b_0^2$, with the uniform-field value reached only on the boundaries $z = \pm L_z/2$. The less obvious lesson concerns the length scale. In the chiral limit there is no intrinsic scale, and the $L_x \to \infty$ form of the solution shows that the disturbance decays as $e^{-\pi|x|/L_z}$. The healing length L_z/π is thus set by the size of the system along the field, not by microscopic physics. This explains why on a square domain the condensate never recovers the uniform-field profile, and why the energy excess of the wall scales as L_z^2 (Appendix A). For fields well above the CSL threshold, the pion mass is only a small perturbation, and the same picture persists, as seen in Fig. 6, where the sharp-wall result stays close to the chiral-limit value 1/2 even for $m_\pi L \gg 1$.

We have verified the statements in (b)–(d) by an independent numerical solution of the chiral-limit problem on the same domains: the identity in (b), the sign pattern, peak radius and magnitude in (c), and the bounds and healing behavior in (d).

Changes in the manuscript: New paragraphs at the end of Sec. 3.1 (frustration mechanism, new Eq. (22) for the baryon number). A new paragraph after the definition of the Gaussian field in Sec. 4.2, with a remark in the caption of Fig. 3 that the overall magnitude is strongly suppressed; we now refer to this configuration as the "ground state" rather than a "CSL solution". A new paragraph after the introduction of Fig. 4 in Sec. 4.3. A new paragraph in Sec. 5 that collects these observations into rules of thumb for the qualitative behavior of the ground state in more general magnetic fields.

Point (4): physical scales

The authors mention in the introduction that inhomogeneities in B are present in heavy-ion collisions and in neutron stars but that at least in the latter variations are on a scale much larger than relevant for modification of the original CSL state. It might be helpful to give some numbers for the scales involved to get an idea how far (if at all) the configurations seen here are away from what we would expect in physical systems. Also, would these estimates change if neutron star mergers are considered, where there are magneto-rotational instabilities and thus small-scale variations of the magnetic field? Presumably even these scales are much larger than what is needed for the configurations considered here, and also temperature effects in mergers would destroy any CSL-type phase?

Reply: We thank the referee for this suggestion and have added a paragraph with order-of-magnitude estimates to the introduction. The structure of the CSL is set by the pion Compton wavelength, $1/m_\pi \approx 1.4$ fm, which is also the scale of the inhomogeneities studied in our paper.

In neutron stars, the magnetic field varies over macroscopic distances, up to ~ 10 km, and stays well below the critical field of $\sim 10^{19}$ G. As the referee anticipated, mergers do not change this conclusion: the Kelvin–Helmholtz and magnetorotational instabilities amplify the field to $\gtrsim 10^{16}$ G, but it still varies on macroscopic scales, and temperatures of several tens of MeV would likely destroy any CSL-type order. Heavy-ion collisions do produce field variations over a few fm, but the field is short-lived and the fireball is too hot. Our configurations are therefore most directly relevant for lattice QCD simulations with nonuniform magnetic fields.

Changes in the manuscript: A new paragraph in the introduction with these estimates and references to the literature on magnetic fields in heavy-ion collisions [1, 2, 3], magnetars [4], field amplification in mergers [5, 6], thermodynamic conditions in mergers [7], and thermal effects on the CSL (references already present in the paper). A sentence on the adiabatic regime of slowly varying fields has also been added to the Summary.

Point (5): Maxwell's equations

Why does the EoM for ϕ not have to be coupled with Maxwell's equations? I understand that the coupling of the pions to B is anomalous, so presumably Maxwell's equations are trivially fulfilled? Even if that's the case maybe a short remark would be helpful.

Reply: We thank the referee for this question. As far as the setup of our paper is concerned, the referee's intuition is correct. We treat $\boldsymbol{B}$ as an external, nondynamical field, created by sources outside the system. For a uniform chemical potential, the anomalous term in the Lagrangian can be written as a total derivative, $\frac{\mu}{4\pi^2}\boldsymbol{B}\cdot\boldsymbol{\nabla}\phi = \frac{\mu}{4\pi^2}\boldsymbol{\nabla}\cdot(\phi\boldsymbol{B})$. Its variation with respect to the vector potential gives the induced electric current

$$\boldsymbol{j} = \frac{1}{4\pi^2}\boldsymbol{\nabla}\times(\mu\boldsymbol{\nabla}\phi) = \frac{1}{4\pi^2}\boldsymbol{\nabla}\mu\times\boldsymbol{\nabla}\phi,$$

which vanishes identically for uniform μ, regardless of the profile of $\boldsymbol{B}$ and ϕ. Maxwell's equations are therefore trivially satisfied by the externally imposed field. The only feedback of the pion field on $\boldsymbol{B}$ comes through the magnetization $\boldsymbol{M} = \mu\boldsymbol{\nabla}\phi/(4\pi^2)$ at the boundary of the system, and it is suppressed by the fine-structure constant (below one percent for $\mu \approx 1$ GeV).

The physics becomes quite different, however, once the electromagnetic field is treated as a dynamical degree of freedom and Maxwell's equations are solved together with the EoM for ϕ. The Wess–Zumino–Witten term then also contributes the coupling of the neutral pion to $\boldsymbol{E}\cdot\boldsymbol{B}$, responsible for the $\pi^0 \to \gamma\gamma$ decay. As a consequence, the neutral pion acts as an axion, and the low-energy electrodynamics of the medium becomes axion electrodynamics [8]. In the CSL background, the two photon polarizations mix with the neutral pion. The resulting spectrum consists of two gapped modes and one gapless mode with a nonrelativistic dispersion relation, whose character depends on the direction of propagation [9]. A similar mixing of photons and neutral pions occurs already in a uniform magnetic field at zero baryon chemical potential, where it qualitatively changes the low-temperature thermodynamics of QCD [10]. The resulting gapless mode can be interpreted as a type-B Nambu–Goldstone mode of generalized global symmetries [11]. With the external magnetic field held fixed, these effects primarily modify the spectrum of fluctuations above the classical ground state, rather than the classical ground state itself, which is the subject of our paper. Extending this analysis to inhomogeneous magnetic fields is an interesting problem, closely related to the study of the excitation spectrum mentioned as a follow-up direction in Sec. 5.

Changes in the manuscript: A new paragraph with Eq. (4) after the definition of $\boldsymbol{b}$ in Sec. 2. It explains why Maxwell's equations are trivially satisfied for an external magnetic field, and it ends with a remark that the physics is qualitatively different for a dynamical electromagnetic field, with references to Refs. [8, 9, 10, 11]. A sentence in Sec. 3.3 refers back to this paragraph for the case of a nonuniform chemical potential.

Other changes

- Corrected "the convolution of the coordinate dependence" to "the product" in Sec. 3.3, since $\boldsymbol{b}$ is the product of μ and $\boldsymbol{B}$.
- Corrected "We leave both of the above equations to the future work" to "... questions to future work" in Sec. 5.

We hope that with these changes, the manuscript is suitable for publication.

Sincerely,
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